

Particle Swarm Optimization (PSO)

* It is a solution which used to find the best solution to a problem.

* It is inspired by how birds fly together in search of food.

Here, each bird represents one possible solution to the problem.

All birds share information with each other to find food faster.

In PSO, these birds are called particles.

In PSO particle is just one possible solution to the problem, each particle remembers 2 things pbest and gbest.

p-best (personal best)

Each particle keeps track of the best position it has visited so far, ~~record~~.

gbest (global best)

Among all the pbest values of all particles gbest is ^{the} one with the best fitness value (best solution overall).

eg:

Considered 5 birds, they searching for food

Bird	Spot it tried	best spot
B1	A, B, C	C
B2	D, E, F	E
B3	G, H, I	I

* After this Birds Share information, and find the gbest.

Bird	Pbest	Food Found
B1	C	40 seeds
B2	E	50 seeds
B3	I	70 seeds

gbest = I, among this Bird 3 Found 70 seeds.

* Both gbest and pbest is used to Find the optimal solution.

$$pbest_i = \arg \max F(x_i(t))$$

$$gbest = \arg \max F(pbest_i)$$

$x_i(t)$: current position of Particle i

* each particle has

1) position $x_i(t)$

2) velocity $v_i(t)$

They are updated as

$$v_i(t+1) = \omega v_i(t) + c_1 r_1 (pbest_i - x_i(t)) + c_2 r_2 (gbest - x_i(t))$$

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

each particle changes its speed and position using these two equations.

Steps

- 1) initialize particles with random positions and velocities.
- 2) Evaluate fitness for each particle.
- 3) update pbest for each particle
- 4) update gbest for whole swarm
- 5) update velocity & position.
- 6) Repeat this until the best solution found.

Applications

- * machine Learning model tuning
- * Feature selection
- * Design optimization.